

```
In [10]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
data = pd.read_csv("C:/Users/gagan/Downloads/data.csv")
print("\nSample ")
print(data.head())
```

Sample

	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	\
0	2014-05-02 00:00:00	313000.0	3.0	1.50	1340	7912	
1	2014-05-02 00:00:00	2384000.0	5.0	2.50	3650	9050	
2	2014-05-02 00:00:00	342000.0	3.0	2.00	1930	11947	
3	2014-05-02 00:00:00	420000.0	3.0	2.25	2000	8030	
4	2014-05-02 00:00:00	550000.0	4.0	2.50	1940	10500	

	floors	waterfront	view	condition	sqft_above	sqft_basement	yr_built	\
0	1.5	0	0	3	1340	0	1955	
1	2.0	0	4	5	3370	280	1921	
2	1.0	0	0	4	1930	0	1966	
3	1.0	0	0	4	1000	1000	1963	
4	1.0	0	0	4	1140	800	1976	

	yr_renovated	street	city	statezip	country
0	2005	18810 Densmore Ave N	Shoreline	WA 98133	USA
1	0	709 W Blaine St	Seattle	WA 98119	USA
2	0	26206-26214 143rd Ave SE	Kent	WA 98042	USA
3	0	857 170th Pl NE	Bellevue	WA 98008	USA
4	1992	9105 170th Ave NE	Redmond	WA 98052	USA

```
In [11]: # Feature selection
X = data[['sqft_living', 'bedrooms', 'bathrooms']]
y = data['price']

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
In [12]: # Train linear regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict and evaluate
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

```
In [13]: print(f'Mean Squared Error: {mse}')
print(f'R² Score: {r2}')
```

Mean Squared Error: 991204834716.4655
R² Score: 0.028084138016367333